TP1 Test Cases

Team 53 | CSE 360 | Phase One. Part 1 lists the input validation tests. They are automated in TP1InputValidationTesting.java, and the IDs match the IDs in that class. Each test compares the value the code returns with the expected result. Part 2 lists feature tests for account deletion and the related interface changes. They are run by hand in the application. Part 3 lists tests for password boundaries, one-time passwords, the dynamic password interface, and the first-user setup.

Part 1: Input Validation Tests

Username (UserNameRecognizer): 20/20 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
UN-01	`abcd`	Shortest valid UserName (4 characters)	Accept	Accept	PASS
UN-02	`[a x 32]`	Longest valid UserName (32 characters)	Accept	Accept	PASS
UN-03	`Josh_99`	Letters, a digit, and a separator are accepted	Accept	Accept	PASS
UN-04	`a.b-c&d_e`	Each of the four separators (. - & _) is accepted	Accept	Accept	PASS
UN-05	`(empty string)`	Empty input is rejected	Reject: The input is empty.	Reject: The input is empty.	PASS
UN-06	`(null)`	Null input is rejected without crashing	Reject: The input is empty.	Reject: The input is empty.	PASS
UN-07	`abc`	UserName one character under the minimum	Reject: A UserName must have at least 4 characters.	Reject: A UserName must have at least 4 characters.	PASS
UN-08	`[a x 33]`	UserName one character over the maximum	Reject: A UserName must have no more than 32 characters.	Reject: A UserName must have no more than 32 characters.	PASS
UN-09	`9lives`	First character must be a letter (digit rejected)	Reject: A UserName must start with A-Z or a-z.	Reject: A UserName must start with A-Z or a-z.	PASS
UN-10	`_josh`	First character must be a letter (separator rejected)	Reject: A UserName must start with A-Z or a-z.	Reject: A UserName must start with A-Z or a-z.	PASS
UN-11	`josh__99`	Two separators in a row are rejected	Reject: A separator must be followed by A-Z, a-z, or 0-9.	Reject: A separator must be followed by A-Z, a-z, or 0-9.	PASS
UN-12	`josh_`	A trailing separator is rejected	Reject: A separator must be followed by A-Z, a-z, or 0-9.	Reject: A separator must be followed by A-Z, a-z, or 0-9.	PASS
UN-13	`josh smith`	A space is not an allowed character	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	PASS
UN-14	`josh@asu`	A disallowed symbol (@) is rejected	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	PASS
UN-15	`jo!`	Illegal character found before the length check runs	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
UN-16	`[a x 31]_`	32 characters ending in a separator	Reject: A separator must be followed by A-Z, a-z, or 0-9.	Reject: A separator must be followed by A-Z, a-z, or 0-9.	PASS
UN-17	`[a x 32]_`	Separator as the 33rd character (too long)	Reject: A UserName must have no more than 32 characters.	Reject: A UserName must have no more than 32 characters.	PASS
UN-18	`josé99`	A non-ASCII letter is rejected	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	PASS
UN-19	`(null)`	Update Username page: null value is rejected before the recognizer runs	Reject: Enter a UserName.	Reject: Enter a UserName.	PASS
UN-20	`josh_99`	Update Username page: a valid value is passed through to the recognizer	Accept	Accept	PASS

Password (Model.evaluatePassword): 16/16 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
PW-01	`Abcdef1!`	Shortest valid password (8 characters, all four character types)	Accept	Accept	PASS
PW-02	`Aa1![x x 16]`	Longest valid password (20 characters)	Accept	Accept	PASS
PW-03	`(empty string)`	Empty input is rejected	Reject: The password is empty!	Reject: The password is empty!	PASS
PW-04	`Aa1![x x 17]`	One character over the maximum (21 characters)	Reject: The password may not exceed 20 characters!	Reject: The password may not exceed 20 characters!	PASS
PW-05	`Abc1!`	Too short: only the minimum-length rule fails	Reject: Long Enough; conditions were not satisfied	Reject: Long Enough; conditions were not satisfied	PASS
PW-06	`Abcdefg1`	Missing special character	Reject: Special character; conditions were not satisfied	Reject: Special character; conditions were not satisfied	PASS
PW-07	`abcdefg1!`	Missing uppercase letter	Reject: Upper case; conditions were not satisfied	Reject: Upper case; conditions were not satisfied	PASS
PW-08	`ABCDEFG1!`	Missing lowercase letter	Reject: Lower case; conditions were not satisfied	Reject: Lower case; conditions were not satisfied	PASS
PW-09	`Abcdefgh!`	Missing digit	Reject: Numeric digits; conditions were not satisfied	Reject: Numeric digits; conditions were not satisfied	PASS
PW-10	`abcdefgh`	Several requirements missing at once are all listed	Reject: Upper case; Numeric digits; Special character; conditions were not satisfied	Reject: Upper case; Numeric digits; Special character; conditions were not satisfied	PASS
PW-11	`Abcdef 1!`	A space is an invalid character (error index points at it)	Reject: An invalid character has been found!	Reject: An invalid character has been found!	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
PW-12	`Abcdéf1!`	A non-ASCII letter is an invalid character	Reject: An invalid character has been found!	Reject: An invalid character has been found!	PASS
PW-13	`Aa1~`!@#$%^&*()_-+={`	Special characters ~ ` ! @ # $ % ^ & * ( ) _ - + = { } are accepted	Accept	Accept	PASS
PW-14	`Aa1}[]|\:;"'<>,.?/`	Special characters [ ] | \ : ; " ' < > , . ? / are accepted	Accept	Accept	PASS
PW-15	`"Ab1 "`	Invalid character reported ahead of the missing-requirement list	Reject: An invalid character has been found!	Reject: An invalid character has been found!	PASS
PW-16	`(null)`	Null input is rejected without crashing	Reject: The password is empty!	Reject: The password is empty!	PASS

Email address (EmailAddressRecognizer): 26/26 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
EM-01	`josh@asu.edu`	A simple valid address	Accept	Accept	PASS
EM-02	`josh.l+asu@mail.asu.edu`	Period and plus in the mailbox name; multi-part domain	Accept	Accept	PASS
EM-03	`a@b.co`	Shortest realistic valid address	Accept	Accept	PASS
EM-04	`(empty string)`	Empty input is rejected	Reject: The email address is empty!	Reject: The email address is empty!	PASS
EM-05	`(null)`	Null input is rejected without crashing	Reject: The email address is empty!	Reject: The email address is empty!	PASS
EM-06	`[a x 64]@[b x 63].[b x 63].[c x 61]`	Longest valid address (254 characters, 64-character mailbox name)	Accept	Accept	PASS
EM-07	`[a x 64]@[b x 63].[b x 63].[c x 62]`	One character over the maximum (255 characters)	Reject: The email address may not exceed 254 characters!	Reject: The email address may not exceed 254 characters!	PASS
EM-08	`joshasu.edu`	Missing the @	Reject: The email address must contain an "@"!	Reject: The email address must contain an "@"!	PASS
EM-09	`josh@@asu.edu`	More than one @	Reject: The email address may contain only one "@"!	Reject: The email address may contain only one "@"!	PASS
EM-10	`@asu.edu`	No mailbox name before the @	Reject: There must be a mailbox name before the "@"!	Reject: There must be a mailbox name before the "@"!	PASS
EM-11	`[a x 65]@asu.edu`	Mailbox name one over the 64-character limit	Reject: The part before the "@" may not exceed 64 characters!	Reject: The part before the "@" may not exceed 64 characters!	PASS
EM-12	`jo sh@asu.edu`	A space in the mailbox name	Reject: The character " " is not allowed before the "@"!	Reject: The character " " is not allowed before the "@"!	PASS
EM-13	`.josh@asu.edu`	Mailbox name starts with a period	Reject: The part before the "@" may not start or end with a period!	Reject: The part before the "@" may not start or end with a period!	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
EM-14	`josh.@asu.edu`	Mailbox name ends with a period	Reject: The part before the "@" may not start or end with a period!	Reject: The part before the "@" may not start or end with a period!	PASS
EM-15	`jo..sh@asu.edu`	Two periods in a row in the mailbox name	Reject: Two periods in a row are not allowed!	Reject: Two periods in a row are not allowed!	PASS
EM-16	`josh@`	No domain after the @	Reject: There must be a domain after the "@"!	Reject: There must be a domain after the "@"!	PASS
EM-17	`josh@asu`	Domain has no period	Reject: The domain must contain a period (e.g., asu.edu)!	Reject: The domain must contain a period (e.g., asu.edu)!	PASS
EM-18	`josh@asu..edu`	Domain has an empty part	Reject: The domain has an empty part; check the periods!	Reject: The domain has an empty part; check the periods!	PASS
EM-19	`josh@-asu.edu`	Domain part starts with a hyphen	Reject: A domain part may not start or end with a hyphen!	Reject: A domain part may not start or end with a hyphen!	PASS
EM-20	`josh@asu-.edu`	Domain part ends with a hyphen	Reject: A domain part may not start or end with a hyphen!	Reject: A domain part may not start or end with a hyphen!	PASS
EM-21	`josh@asu.e`	Top-level domain is only one letter	Reject: The domain must end with at least two letters (e.g., .edu)!	Reject: The domain must end with at least two letters (e.g., .edu)!	PASS
EM-22	`josh@asu.e1`	Top-level domain contains a digit	Reject: The domain must end with letters only (e.g., .edu)!	Reject: The domain must end with letters only (e.g., .edu)!	PASS
EM-23	`josh@as_u.edu`	Illegal character in the domain	Reject: The character "_" is not allowed in the domain!	Reject: The character "_" is not allowed in the domain!	PASS
EM-24	`josh@asu.edu.`	Trailing period after the domain	Reject: The domain has an empty part; check the periods!	Reject: The domain has an empty part; check the periods!	PASS
EM-25	`jösh@asu.edu`	A non-ASCII letter in the mailbox name	Reject: The character "ö" is not allowed before the "@"!	Reject: The character "ö" is not allowed before the "@"!	PASS
EM-26	`(null)`	Update Email Address page: null value is rejected before the recognizer runs	Reject: Enter an email address.	Reject: Enter an email address.	PASS

Names: First, Middle, Last, Preferred First (InputValidation.checkName): 17/17 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
NM-01	`Joshua`	An ordinary name is accepted	Accept	Accept	PASS
NM-02	`(empty string)`	Empty value is accepted (clears the field)	Accept	Accept	PASS
NM-03	`(null)`	Null (dialog cancelled) is accepted and changes nothing	Accept	Accept	PASS
NM-04	`Mary-Jane O'Neil`	Hyphen, space, and apostrophe are accepted	Accept	Accept	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
NM-05	`[a x 50]`	Longest valid name (50 characters)	Accept	Accept	PASS
NM-06	`[a x 51]`	One character over the maximum (51 characters)	Reject: A name may not exceed 50 characters.	Reject: A name may not exceed 50 characters.	PASS
NM-07	`Josh3`	A digit is rejected	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "3" is not allowed.	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "3" is not allowed.	PASS
NM-08	`Josh_`	An underscore is rejected	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "_" is not allowed.	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "_" is not allowed.	PASS
NM-09	`Jos.h`	A period is rejected	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "." is not allowed.	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "." is not allowed.	PASS
NM-10	`José`	An accented letter is accepted	Accept	Accept	PASS
NM-11	`" " (3 spaces)`	Only spaces is rejected (no letters)	Reject: A name must contain at least one letter.	Reject: A name must contain at least one letter.	PASS
NM-12	`-`	Only a hyphen is rejected (no letters)	Reject: A name must contain at least one letter.	Reject: A name must contain at least one letter.	PASS
NM-13	`[a x 51]3`	Length check runs before the character check	Reject: A name may not exceed 50 characters.	Reject: A name may not exceed 50 characters.	PASS
NM-14	`'`	Only an apostrophe is rejected (no letters)	Reject: A name must contain at least one letter.	Reject: A name must contain at least one letter.	PASS
NM-15	`O'Brien-Smith Jr`	Letters mixed with apostrophe, hyphen, and space are accepted	Accept	Accept	PASS
NM-16	`Zoë Ångström`	Accented and other non-ASCII letters are accepted	Accept	Accept	PASS
NM-17	`José3`	A digit is still rejected after an accented letter	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "3" is not allowed.	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "3" is not allowed.	PASS

Invitation role (InputValidation.validateRole): 9/9 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
RL-01	`Admin`	Admin is an accepted role	Accept	Accept	PASS
RL-02	`Contributor`	Contributor is an accepted role	Accept	Accept	PASS
RL-03	`Viewer`	Viewer is an accepted role	Accept	Accept	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
RL-04	`Curator`	Curator is an accepted role	Accept	Accept	PASS
RL-05	`(null)`	No role chosen (null)	Reject: Select a role for the invitation.	Reject: Select a role for the invitation.	PASS
RL-06	`Select a role`	The placeholder choice is not a role	Reject: Select a role for the invitation.	Reject: Select a role for the invitation.	PASS
RL-07	`Superuser`	A role that does not exist is rejected	Reject: The role "Superuser" is not one of the available roles.	Reject: The role "Superuser" is not one of the available roles.	PASS
RL-08	`admin`	Role names are case sensitive	Reject: The role "admin" is not one of the available roles.	Reject: The role "admin" is not one of the available roles.	PASS
RL-09	`(empty string)`	An empty role is rejected	Reject: The role "" is not one of the available roles.	Reject: The role "" is not one of the available roles.	PASS

Invitation code (InputValidation.validInvitationInput): 8/8 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
IC-01	`ABC123`	A typical code is accepted	Accept	Accept	PASS
IC-02	`A`	Shortest valid code (1 character)	Accept	Accept	PASS
IC-03	`aaaaaaaaaa`	Longest valid code (10 characters)	Accept	Accept	PASS
IC-04	`aaaaaaaaaaa`	One character over the maximum	Reject: enter a code of 1 to 10 characters	Reject: enter a code of 1 to 10 characters	PASS
IC-05	`(empty string)`	Empty input is rejected	Reject: enter a code of 1 to 10 characters	Reject: enter a code of 1 to 10 characters	PASS
IC-06	`" " (3 spaces)`	Whitespace-only input is rejected	Reject: enter a code of 1 to 10 characters	Reject: enter a code of 1 to 10 characters	PASS
IC-07	`(null)`	Null input is rejected	Reject: enter a code of 1 to 10 characters	Reject: enter a code of 1 to 10 characters	PASS
IC-08	`" ABC "`	Spaces around a code are not trimmed; 5 characters is within the limit	Accept	Accept	PASS

Invitation deadline time (InputValidation.parseInvitationDeadline): 13/13 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
TM-01	`2026-10-01, 09:30`	A valid time is combined with the date	Accept: 2026-10-01 09:30	Accept: 2026-10-01 09:30	PASS
TM-02	`2026-10-01, 17:05`	A valid afternoon time (24-hour clock)	Accept: 2026-10-01 17:05	Accept: 2026-10-01 17:05	PASS
TM-03	`2026-10-01, 00:00`	Lower boundary of the day	Accept: 2026-10-01 00:00	Accept: 2026-10-01 00:00	PASS
TM-04	`2026-10-01, 23:59`	Upper boundary of the day	Accept: 2026-10-01 23:59	Accept: 2026-10-01 23:59	PASS
TM-05	`2026-10-01, " 09:30 "`	Leading and trailing spaces are ignored	Accept: 2026-10-01 09:30	Accept: 2026-10-01 09:30	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
TM-06	`2026-10-01, 9:30`	Hour must have two digits	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-07	`2026-10-01, 24:00`	Hour 24 is not a valid time	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-08	`2026-10-01, 12:60`	Minute 60 is not a valid time	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-09	`2026-10-01, ab:cd`	Non-numeric time	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-10	`2026-10-01, (empty)`	Empty time	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-11	`2026-10-01, 09:30:00`	Longer than 5 characters after trimming is rejected	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-12	`(no date), 09:30`	No date chosen	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-13	`2026-10-01, (no time)`	No time entered	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS

Invitation deadline policy (InputValidation.validateInvitationDeadline): 6/6 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
DL-01	`(cannot be parsed)`	Deadline could not be parsed	Reject: Enter a valid expiration date and a time in the form HH:mm (24-hour clock).	Reject: Enter a valid expiration date and a time in the form HH:mm (24-hour clock).	PASS
DL-02	`now + 1 day`	One day from now is accepted	Accept	Accept	PASS
DL-03	`now - 1 minute`	One minute ago is rejected	Reject: The expiration date and time must be in the future.	Reject: The expiration date and time must be in the future.	PASS
DL-04	`now + 30 days - 1 minute`	Just inside the 30-day limit is accepted	Accept	Accept	PASS
DL-05	`now + 30 days + 1 hour`	Just past the 30-day limit is rejected	Reject: The expiration date and time must be within 30 days from now.	Reject: The expiration date and time must be within 30 days from now.	PASS
DL-06	`now + 31 days`	Well past the limit is rejected	Reject: The expiration date and time must be within 30 days from now.	Reject: The expiration date and time must be within 30 days from now.	PASS

Issues found by the input validation tests and fixed

Issue	Fix
Password check crashed on null input (PW-16)	Null is now rejected as an empty password.
Names of only spaces, a hyphen, or an apostrophe were accepted (NM-11, NM-12, NM-14)	A name must contain at least one letter.
Accented letters were rejected in names (NM-10, NM-16)	Any letter is now accepted.
The time 24:00 was accepted as midnight of the same day (TM-07)	The hour must be 00 to 23.
The invitation role defaulted to Admin (RL-05 to RL-09)	The drop-down starts on "Select a role", and a role must be chosen.
After a one-time password, the user was not sent back to Log In	The user now returns to the Log In page.
The name error alert always said "Name"	The alert now names the specific field.

Part 2: Feature Tests (account deletion and interface)

These cases cover account deletion and the related interface changes. Use disposable accounts for deletion checks. Each case lists the action and the expected result.

#	Test Case	Action	Expected result
FT-01	Cancel deletion	Select an account and choose No in the confirmation message.	The account remains available.
FT-02	Close the confirmation	Open the confirmation and press Escape or close the message.	The account is not deleted.
FT-03	No account selected	Open the deletion page without selecting an account.	The Delete account button is disabled.
FT-04	Confirm deletion	Select another account and choose Yes.	The selected account is removed and the list is refreshed.
FT-05	Delete own account	Select the signed-in administrator account and confirm.	The app refuses to delete the account and explains why.
FT-06	Check administrator access	Try the deletion operation without an administrator session.	The operation is refused and no accounts are changed.
FT-07	Check deleted account access	After deletion, try the removed account's normal password and any active one-time password.	Neither gives access to the removed account.
FT-08	Check the selected account	Select an account and read the details and confirmation message.	The name, email and roles match the selection. The confirmation identifies the account to be deleted.
FT-09	Check an old selection	Remove an account, create another with the same username, and try deleting through the old selection.	The replacement account is not deleted.
FT-10	Check a failed database operation	Use a test database that prevents deletion and attempt to remove an account.	The operation reports failure and the account remains.
FT-11	Check changed permissions	Remove the signed-in user's administrator role before they attempt deletion.	Deletion is refused.
FT-12	Check renamed administrator	Change the signed-in administrator's username, then try to delete that account.	Self-deletion is still refused.
FT-13	Check invitation input	Try a blank invitation code and one longer than 10 characters.	The app asks for a code between 1 and 10 characters. A code within that length still needs to be a valid invitation.
FT-14	Check login input	Try a username longer than 32 characters or a password longer than 64 characters.	The login attempt is rejected.

#	Test Case	Action	Expected result
FT-15	Check the confirmation buttons	Open the deletion confirmation and inspect the choices.	Yes is not the default action. No is the default and cancel action.
FT-16	Check the shared appearance	Open login and deletion and move between the application pages.	The shared style is applied, the fields are readable and the buttons remain usable.

Part 3: Password Reset, One-Time Password, and First-User Tests

These cases cover input validation and password boundaries, one-time passwords and password reset, the dynamic password interface, and the fresh login after first-user setup. The Verified column records the result of each case.

3A. Input Validation and Password Boundaries: 46/46 test cases passed

ID	How tested (data / action)	Expected result	Verified
IV-01	Validate username agupt545.	Accepted as a valid username.	PASS
IV-02	Validate an empty username.	Refused; username cannot be empty.	PASS
IV-03	Validate username 1abc.	Refused because it begins with a digit.	PASS
IV-04	Paste a 10,000-character username.	Refused safely without delay or failure.	PASS
IV-05	Validate password Aa!15678.	Accepted; all required character categories are present.	PASS
IV-06	Validate password aa!15678 with no uppercase letter.	Refused; uppercase requirement remains unmet.	PASS
IV-07	Validate password Aa115678 with no special character.	Refused; special-character requirement remains unmet.	PASS
IV-08	Validate the 7-character password Aa!1567.	Refused because it is shorter than 8 characters.	PASS
IV-09	Validate a password exactly 20 characters long.	Accepted when all composition rules are satisfied.	PASS
IV-10	Validate a password 21 characters long.	Refused because it exceeds the 20-character maximum.	PASS
IV-11	Paste a 10,000-character password.	Refused safely; no stale valid result remains.	PASS
IV-12	Validate email agupt545@asu.edu.	Accepted as a valid email address.	PASS
IV-13	Validate first.last+tag_1%x-y@sub.example.co.uk.	Accepted; supported mailbox and subdomain punctuation works.	PASS
IV-14	Validate an empty email address.	Refused; email cannot be empty.	PASS
IV-15	Validate an email with no @ symbol.	Refused.	PASS
IV-16	Validate an email containing two @ symbols.	Refused.	PASS
IV-17	Validate an email with no mailbox before @.	Refused.	PASS
IV-18	Validate an email with no domain after @.	Refused.	PASS
IV-19	Validate a domain with no period or top-level domain.	Refused.	PASS
IV-20	Validate an email with a one-letter top-level domain.	Refused.	PASS
IV-21	Validate a mailbox containing consecutive periods.	Refused.	PASS
IV-22	Validate a mailbox ending in a period.	Refused.	PASS
IV-23	Validate a domain label beginning with a hyphen.	Refused.	PASS
IV-24	Validate an email containing an embedded space.	Refused.	PASS

ID	How tested (data / action)	Expected result	Verified
IV-25	Validate an email longer than the documented maximum.	Refused at the length boundary.	PASS
IV-26	Paste a 10,000-character email.	Refused safely without delay or failure.	PASS
IV-27	Run a failing recognizer and inspect its error index.	The recognizer records the location of the first error.	PASS
IV-28	Validate name Aditya.	Accepted as a valid name.	PASS
IV-29	Validate name Anne-Marie O'Brien.	Accepted; hyphen and apostrophe are supported.	PASS
IV-30	Clear the optional name field.	Accepted so the field can be cleared.	PASS
IV-31	Validate a name containing a digit.	Refused.	PASS
IV-32	Validate a name exactly at its maximum length.	Accepted.	PASS
IV-33	Validate a name one character over its maximum length.	Refused.	PASS
IV-34	Paste a 10,000-character name.	Refused safely without delay or failure.	PASS
IV-35	Update an account using the shared username validator.	The account-update path applies the same username rules.	PASS
IV-36	Update an account using the shared email validator.	The account-update path applies the same email rules.	PASS
IV-37	Pass a null value to password validation.	Refused safely; no exception is thrown.	PASS
IV-38	Enter a password containing a tab character.	Refused as invalid whitespace.	PASS
IV-39	Validate a fully compliant password and inspect requirement flags.	All observable requirement flags become satisfied.	PASS
IV-40	Validate a good password, then paste an overlong value.	All earlier success flags are cleared.	PASS
IV-41	Validate a good password, then clear the field.	All earlier success flags are cleared.	PASS
IV-42	Paste an excessive password, then enter a valid password.	Validation recovers and the valid password succeeds.	PASS
IV-43	Confirm a password exactly 20 characters long.	Accepted at the maximum boundary.	PASS
IV-44	Confirm a password 21 characters long.	Refused above the maximum boundary.	PASS
IV-45	Paste 10,000 characters into password confirmation.	Refused safely without delay or failure.	PASS
IV-46	Pass null to password confirmation.	Refused safely; no exception is thrown.	PASS

3B. One-Time Password and Password Reset: 33/33 test cases passed

ID	How tested (data / action)	Expected result	Verified
OTP-01	Create a new user and query one-time-password status.	The user begins with no active one-time password.	PASS
OTP-02	Generate 100 one-time passwords.	Every generated value satisfies the permanent-password rules.	PASS
OTP-03	Inspect the generated one-time-password length.	Every generated value has the documented length of 12.	PASS
OTP-04	Generate two one-time passwords for comparison.	The generated values differ.	PASS
OTP-05	Store a one-time password with a future deadline.	The credential is stored successfully.	PASS
OTP-06	Query status after storing a future-deadline credential.	The credential is reported active.	PASS
OTP-07	Read the stored expiration deadline.	The stored deadline matches the requested deadline.	PASS

ID	How tested (data / action)	Expected result	Verified
OTP-08	Attempt login with the wrong one-time password.	Login is refused.	PASS
OTP-09	After a wrong attempt, query the original credential.	The valid one-time password remains active.	PASS
OTP-10	Use the correct one-time password before its deadline.	Login is accepted.	PASS
OTP-11	Immediately query the credential after successful login.	The one-time password has been consumed.	PASS
OTP-12	Try to reuse the successfully redeemed credential.	Reuse is refused.	PASS
OTP-13	Inspect the permanent password immediately after OTP login.	It is unchanged until the required reset is completed.	PASS
OTP-14	Submit a valid replacement permanent password.	The reset completes successfully.	PASS
OTP-15	Read the account after reset.	The new permanent password is stored.	PASS
OTP-16	Start a fresh session and use the new permanent password.	The new password is accepted.	PASS
OTP-17	Try the old permanent password after reset.	The old password is refused.	PASS
OTP-18	Attempt to assign a one-time password with a past deadline.	The assignment is refused.	PASS
OTP-19	Attempt to assign a one-time password without a deadline.	The assignment is refused.	PASS
OTP-20	Attempt to assign an empty one-time password.	The assignment is refused.	PASS
OTP-21	Attempt to assign a one-time password to a missing user.	The assignment is refused.	PASS
OTP-22	Attempt login using an expired one-time password.	Login is refused.	PASS
OTP-23	Query status after the expired credential is detected.	The expired credential is purged.	PASS
OTP-24	Submit a 10,000-character one-time-password value.	The input is refused safely.	PASS
OTP-25	After the oversized attempt, try the real credential.	The oversized input did not consume the valid credential.	PASS
OTP-26	Submit null as the one-time-password input.	The input is refused safely; no exception is thrown.	PASS
OTP-27	After the null attempt, try the real credential.	The null input did not consume the valid credential.	PASS
OTP-28	Replace an active one-time password and try the old value.	The replaced value no longer works.	PASS
OTP-29	After trying the old value, query the replacement.	The replacement remains active.	PASS
OTP-30	Redeem the replacement one-time password.	The replacement is accepted.	PASS
OTP-31	Try to reuse the redeemed replacement.	The replacement is consumed after its first successful use.	PASS
OTP-32	Start two simultaneous redemptions of the same credential.	Exactly one redemption succeeds.	PASS
OTP-33	Query the credential after the concurrent attempts.	The concurrently redeemed credential is no longer active.	PASS

3C. Dynamic Password Interface: 6/6 test cases passed

ID	How tested (data / action)	Expected result	Verified
DPV-P-01	Open the password-reset window and inspect the entry field.	Password entry is masked.	PASS
DPV-P-02	Enter a password that satisfies all live requirements.	The Finish button becomes enabled.	PASS
DPV-P-03	Erase a previously valid password.	The Finish button becomes disabled immediately.	PASS

ID	How tested (data / action)	Expected result	Verified
DPV-P-04	After valid input, paste an excessive-length password.	Stale valid state is cleared and Finish is disabled.	PASS
DPV-P-05	Close and reopen the password-reset window.	The field starts empty and Finish starts disabled.	PASS
DPV-P-06	Trigger a validation error and inspect the displayed message.	The error marker never echoes the password text.	PASS

3D. Fresh Login After First-User Setup: 4/4 test cases passed

ID	How tested (data / action)	Expected result	Verified
FUR-R-01	Complete first-user setup and inspect the completion message.	The message clearly requires the user to log in again.	PASS
FUR-R-02	Finish first-user setup and inspect the displayed scene.	The application returns to the login scene.	PASS
FUR-R-03	Finish first-user setup and inspect session state.	The authenticated session is cleared.	PASS
FUR-R-04	Display the ordinary user home after the first-user flow.	The singleton view resets to the normal role-home action.	PASS